10. Overall results

We will now summarize the global characteristics of the different datasets and the classification performance of the Kmeans algorithm on each of them.

10.1. Datasets used

- `DBT2/NDBT2 dataset` ($n = 39$) with 19 DBT2 patients and 20 NDBT2 patients, average 1200 points per patient, 15-minute sampling frequency.
- `CGMacros` ($n = 29$) with 14 DBT2 patients and 15 NDBT2 patients, average 1000 points per patient, resampled to 15-minute sampling frequency. The measurements were taken with two different probes: dexcom and libre.
- `Hall` ($n = 43$) with 5 DBT2 patients and 38 NDBT2 patients, average 700 points per patient, resampled to 15-minute sampling frequency.

10.2. Permutation entropy and kmeans parameters

Following our preliminary study, we computed the regular Shannon permutation entropy and MPR complexity with the parameters **(m=5, τ=2)** for all datasets.

To classified the patients, we showed that applying the Kmeans algorithm on the HxC feature plane with the following parameters was able to achieve good classification performance:

- Distance metric: `cosine`
- Maximum of Iteration: 100
- Replicates: 1
- Starting cluster centroid: `plus`

10.3. Classification results

Dataset	Normalized permutation entropy (H)	MPR complexity (C)
DBT2	0.9518	0.0791
CGMacros + Libre DBT2	0.9155	0.1147
CGMacros + Dexcom DBT2	0.9155	0.1147
Hall DBT2	0.9363	0.1071
NDBT2	0.8303	0.2189
CGMacros + Libre NDBT2	0.8171	0.2201
CGMacros + Dexcom NDBT2	0.8171	0.2201
Hall NDBT2	0.8580	0.1985

Table 15: Normalized permutation entropy and MPR complexity for the centroids of each group of patients in the different datasets

10.4. Classification performance

With optimized permutation entropy and kmeans algortihm parameters, we obtained the following classification performance inside each dataset:

Group	Balanced Accuracy	F1-score	Sensibility	Specificity
DBT2/NDBT2	0.98	0.98	0.95	0.91
CGMacros + Dexcom	0.79	0.81	0.87	0.71
CGMacros + Libre	0.90	0.90	0.87	0.93
Average CGMacros	0.845	0.855	0.87	0.82
DBT2/NDBT2 + CGMacros	0.93	0.93	0.92	0.94
Hall	0.85	0.93	0.89	0.80

Table 16: Classification performance of the Kmeans algorithm applied on the HxC feature plane of the different datasets

Then, we tested the centroids of given datasets to classify the other datasets:

Group	Centroids source	Balanced Accuracy	F1-score	Sensibility	Specificity
CGMacros + Dexcom	DBT2/ NDBT2	0.83	0.81	0.73	0.93
CGMacros + Libre	DBT2/ NDBT2	0.93	0.93	0.87	1.00
Hall	DBT2/ NDBT2 + CGMacros	0.86	0.95	0.92	0.80

Table 17: Classification performance of the Kmeans algorithm applied on the HxC feature plane of the different datasets using the centroids found with the Kmeans algorithm on other datasets